

Personal PDF Converter System

Technology Stack

- Backend: Python + Django
- Frontend: Django Templates + Tailwind CSS
- PDF Tools: LibreOffice Headless, PyPDF2, ReportLab, pdfplumber, pytesseract (OCR)

Phase 1 (MVP)

- Word → PDF
- Excel → PDF
- Download PDF
- Drag & Drop Upload
- Preview Before Download
- Dark Mode

Phase 2

- Merge PDF
- Split PDF
- Compress PDF
- Rotate PDF
- Recent Conversion History

Phase 3

- OCR (Scan PDF/Image to Text)
- Smart Table Formatting
- Auto Landscape for Wide Excel
- Batch Conversion
- Keyboard Shortcuts

Important Features

- Prevent table breaking
- Preserve original layout
- Auto fit content width
- Multi-page table support
- Keep fonts and images

System Flow

- Upload File → Django View → LibreOffice/PDF Service → Generate PDF → Preview → Download

Project Structure

- converter/templates/
- converter/static/
- converter/services/
- views.py
- urls.py
- forms.py

Tailwind CSS UI

- Responsive layout
- Drag & Drop zone
- Modern cards
- Sidebar navigation
- Dark mode
- Loading animation
- Progress bar
- File preview

Recommended Development Order

- 1. Django Setup
- 2. Tailwind CSS Setup
- 3. File Upload
- 4. Word → PDF
- 5. Excel → PDF
- 6. Preview & Download
- 7. Merge PDF
- 8. Split PDF
- 9. Compress PDF
- 10. OCR